Structural Selection in the String Landscape

A MAAT-Based Phenomenological Framework for Vacuum Ranking

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Abstract

String theory naturally leads to large spaces of admissible backgrounds, compactifications, flux choices, and effective vacua. This paper proposes a phenomenological structural-selection layer over such candidate backgrounds. The proposal does not modify the string-theoretic equations of motion and does not claim to solve the landscape problem. Its purpose is narrower: to define an explicit ranking architecture for admissible or approximately admissible string backgrounds.

The central object is a MAAT landscape functional

$$F_{\mathrm{MAAT}}^{\mathrm{landscape}}[X] = -\sum_{a\in\{H,B,S,V,R\}} \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad \Gamma_a[X] = \frac{1}{1+d_a[X]},$$

where the five sectors measure equation consistency, balance constraints, controlled activity, inter-sector connectivity, and robustness. The resulting structural measure weights string backgrounds by

$$P_{\mathrm{MAAT}}[X] \propto \exp[-F_{\mathrm{MAAT}}^{\mathrm{landscape}}[X]].$$

The framework is illustrated using existing reproducible toy and bridge benchmarks: a 10D-inspired flux/tadpole ensemble, KKLT-style effective stationary points, period-controlled flux backgrounds, and phenomenological backreaction and Standard-Model-sector proxies. These benchmarks show that structural ranking is not identical to energy-only ordering. They are not a full compactification scan, not a Standard Model construction, and not a unique string measure.

1 Status and Scope

This paper is a phenomenological framework paper. It should be read as a proposal for ranking candidate string backgrounds, not as a derivation of string theory, a proof of a unique vacuum measure, or a solution of the cosmological-constant problem. The guiding separation is:

> *String theory defines the candidate landscape. MAAT defines a structural ranking over that landscape.*

The aim is therefore complementary to string dynamics. The string action, supergravity equations, compactification constraints, brane/orientifold content, moduli dynamics, and flux quantisation define what is admissible. The MAAT layer asks which admissible candidates are structurally favoured once equation residuals, tadpole closure, stability, couplings, and phenomenological obstructions are evaluated.

2 Configuration Space

Let X denote a candidate string background. In the broadest phenomenological notation,

$$X = (g_{10}, \mathcal{M}_6, F_p, \tau, \phi_i, \text{branes}, \text{orientifolds}, \text{moduli}) \,, \tag{1}$$

where g_{10} is the ten-dimensional metric, $\mathcal{M}_6$ is the compact internal space, F_p are fluxes, τ denotes the axio-dilaton or effective modulus sector, and ϕ_i denotes additional matter or moduli fields. The notation is intentionally schematic: different string constructions will instantiate X differently.

The corresponding unselected reference measure is written as

$$\mathrm{d}\mu_{\text{string}}[X], \tag{2}$$

which may represent a flux ensemble, a compactification scan, a discretised toy landscape, or a weighted ensemble of effective vacua. The reference measure is part of the definition of the ranking problem. Without it, there is no well-defined candidate set to select from.

3 MAAT String Master Formula

The proposed structural measure over the string landscape is

$$\boxed{Z^{\text{string}}_{\text{MAAT}}[J] = \int \mathrm{d}\mu_{\text{string}}[X] \, \exp\left[\frac{i}{\hbar} S_{\text{string}}[X] - \beta F^{\text{landscape}}_{\text{MAAT}}[X] + J \cdot \mathcal{O}[X]\right].} \tag{3}$$

The string-theoretic dynamics remain in S_{string}. The new ingredient is the structural-selection functional

$$\boxed{F^{\text{landscape}}_{\text{MAAT}}[X] = -\sum_{a \in \{H,B,S,V,R\}} \lambda_a \log(\epsilon + \Gamma_a[X]) \,, \qquad \Gamma_a[X] = \frac{1}{1 + d_a[X]}.} \tag{4}$$

Here $d_a[X] \geq 0$ are dimensionless sector defects, $\Gamma_a[X] \in (0, 1]$ are sector supports, $\lambda_a > 0$ are sector weights, and $\epsilon > 0$ is a regularisation floor.

The induced selection probability is

$$P_{\text{MAAT}}[X] = \frac{1}{Z} \exp[-F^{\text{landscape}}_{\text{MAAT}}[X]]. \tag{5}$$

The preferred candidates are therefore those that minimise structural defects, not necessarily those that minimise a single energy proxy.

4 Five Structural Sectors in String Theory

The MAAT sectors are interpreted as diagnostic sectors over candidate string backgrounds:

Table 1: String-landscape interpretation of the five MAAT sectors.

Sector	Name	String-landscape meaning
H	Consistency	Equation residuals: ten-dimensional field equations, effective equations of motion, stationarity of the potential.
B	Balance	Tadpole cancellation, anomaly cancellation, flux balance, Bianchi identities, charge closure.
S	Activity	Controlled nontrivial structure: flux activity, moduli stabilisation, hierarchy generation, non-degenerate dynamics.
V	Connectivity	Coupling coherence between geometry, fluxes, branes, moduli, visible-sector proxies, and hidden sectors.
R	Robustness	Stability under perturbations: no tachyonic directions, resistance to runaway, barrier structure, backreaction control.

This table does not define a unique microscopic string measure. It defines a diagnostic basis that can be instantiated in increasingly realistic models.

5 Primitive Defect Closure

A minimal phenomenological closure is

$$d_H[X] = \mathcal{N}_H \left[\left\| \frac{\delta S_{\text{string}}}{\delta X} \right\|^2 + \|\nabla V_{\text{eff}}\|^2 \right], \tag{6}$$

$$d_B[X] = \mathcal{N}_B \left[|\Delta Q_{\text{tadpole}}|^2 + \|\mathrm{d}F - H \wedge F + \text{sources}\|^2 + \mathcal{A}^2_{\text{anom}} \right], \tag{7}$$

$$d_S[X] = \mathcal{N}_S[\text{activity mismatch, hierarchy defect, or moduli-stabilisation defect}], \tag{8}$$

$$d_V[X] = \mathcal{N}_V[\text{coupling mismatch, sector decoupling, or correlation defect}], \tag{9}$$

$$d_R[X] = \mathcal{N}_R[\text{tachyon defect, runaway defect, barrier defect, or backreaction defect}]. \tag{10}$$

The maps $\mathcal{N}_a$ are dimensionless normalisations. In a full compactification scan they should be fixed by physical scales, discretisation scales, data uncertainties, or ensemble statistics.

For benchmark calibration one may use the linear proxy

$$F^{\text{landscape}}_{\text{MAAT}}[X] \approx \sum_a \lambda_a d_a[X], \tag{11}$$

but the logarithmic support form in Eq. (4) is the canonical selection functional because it treats each sector as a multiplicative support factor.

6 Relation to Existing Landscape Ideas

The proposed layer is distinct from, but compatible with, existing string-landscape concepts:

- Flux statistics count or sample large classes of vacua; MAAT adds a structural ranking over sampled candidates.
- Swampland criteria define admissibility or exclusion constraints; MAAT can include such constraints as hard penalties or balance/robustness defects.

- Anthropic reasoning conditions on observer-compatible quantities; MAAT is intended as a pre-anthropic structural prior.
- KKLT-type constructions stabilise moduli in specific effective potentials; MAAT ranks resulting candidates by a multi-sector diagnostic.
- No-boundary or cosmological wavefunction proposals define cosmological weights; MAAT defines structural supports over candidate backgrounds.

The strongest conservative statement is:

> *MAAT is not a replacement for the string landscape. It is a proposed structural ranking functional over that landscape.*

7 Benchmark Evidence

The repository contains a sequence of toy and bridge tests implementing this logic at increasing levels of structure. They are intentionally limited, but they test whether structural ranking contains information beyond energy-only ordering.

7.1 10D-Inspired Tadpole Toy Ensemble

The first benchmark samples 320 flux candidates with a D3-tadpole closure proxy. Energy ranking and structural ranking select substantially different top candidates. Among the top 20 candidates, the overlap between energy ranking and structural ranking is only 6 candidates. The structural top-20 subset has a lower mean tadpole mismatch:

$$\langle|\Delta Q|\rangle_{\text{struct}} = 1.9, \qquad \langle|\Delta Q|\rangle_{\text{energy}} = 4.4.$$

The phenomenological ranking improves this further:

$$\langle|\Delta Q|\rangle_{\text{pheno}} = 1.35.$$

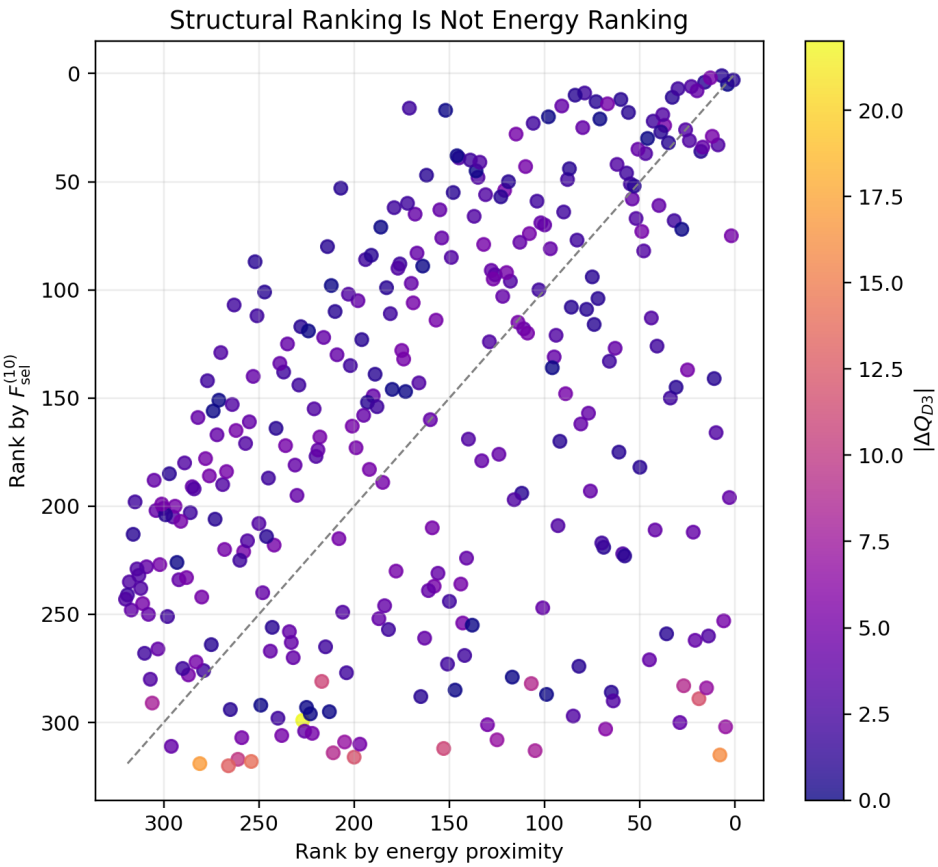

Figure 1: 10D-inspired tadpole toy ensemble. Energy-only ranking and structural ranking select different candidates; structural ranking favours lower tadpole obstruction on average.

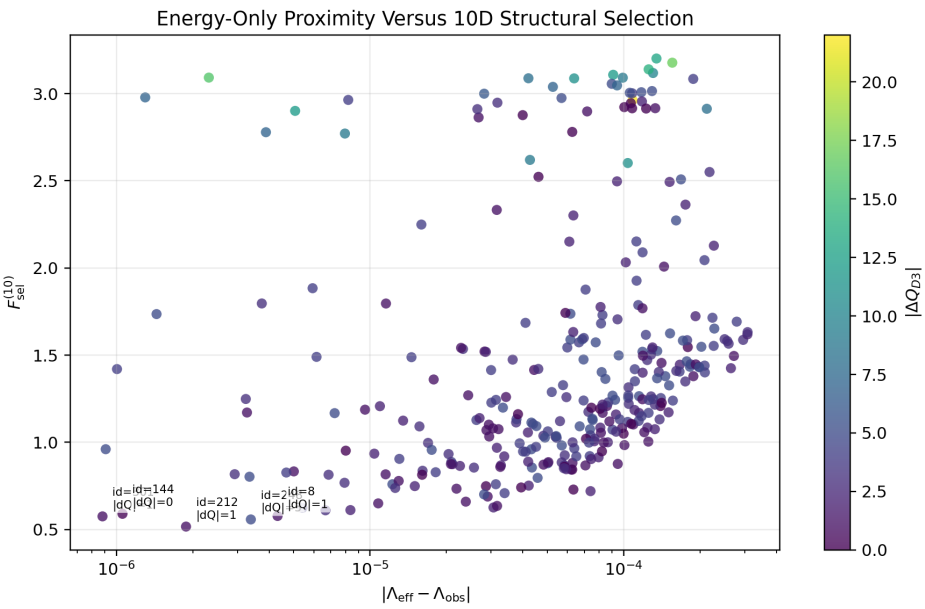

Figure 2: Energy proxy versus structural score in the tadpole toy ensemble. The scatter illustrates that structural selection is not identical to energy-only ordering.

7.2 KKLT and Period-Bridge Benchmarks

The next benchmarks connect flux data to KKLT-style effective stationary points. In the reduced KKLT bridge, the top energy subset has mean $|\Delta Q| = 3.17$, while the phenomenological structural subset has mean $|\Delta Q| = 0.42$. In the exact-period bridge, the top energy subset has mean $|\Delta Q| = 4.67$, while the exact-period structural subset has mean $|\Delta Q| = 0.50$.

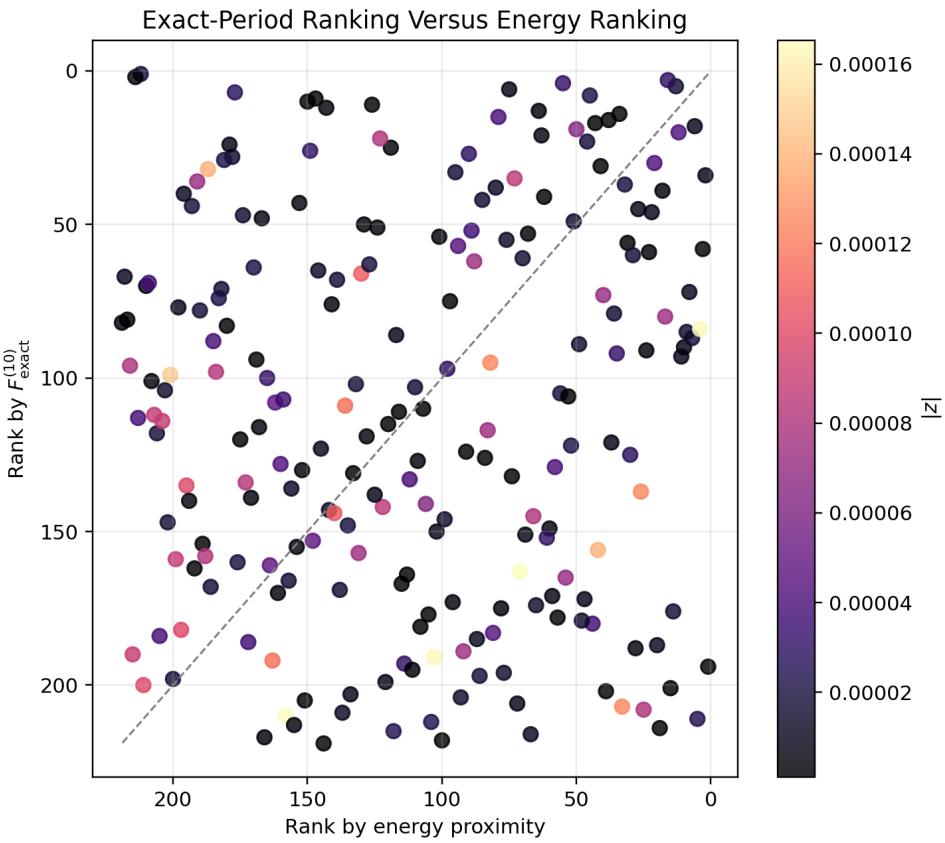

Figure 3: Exact-period-to-KKLT bridge benchmark. Structural ranking separates candidate quality from energy-only ordering and favours lower tadpole obstruction in the tested ensemble.

7.3 Backreaction and Standard-Model-Sector Proxies

The final bridge adds phenomenological backreaction and Standard-Model-sector proxies. The dataset contains 90 backgrounds and 172 stationary candidates, with 47 dS minima, 34 AdS minima, 90 unstable stationary points, and one near-Minkowski minimum. The top-12 overlap between energy ranking and full phenomenological ranking is only one candidate.

Energy ranking gives lower mean absolute energy by construction, but the full structural ranking improves several structural proxies:

$$\langle d_{\text{SM}} \rangle_{\text{energy}} = 1.485, \qquad \langle d_{\text{SM}} \rangle_{\text{full}} = 0.886.$$

The backreaction-SM ranking gives an even lower mean SM-sector defect:

$$\langle d_{\text{SM}} \rangle_{\text{backreaction+SM}} = 0.733.$$

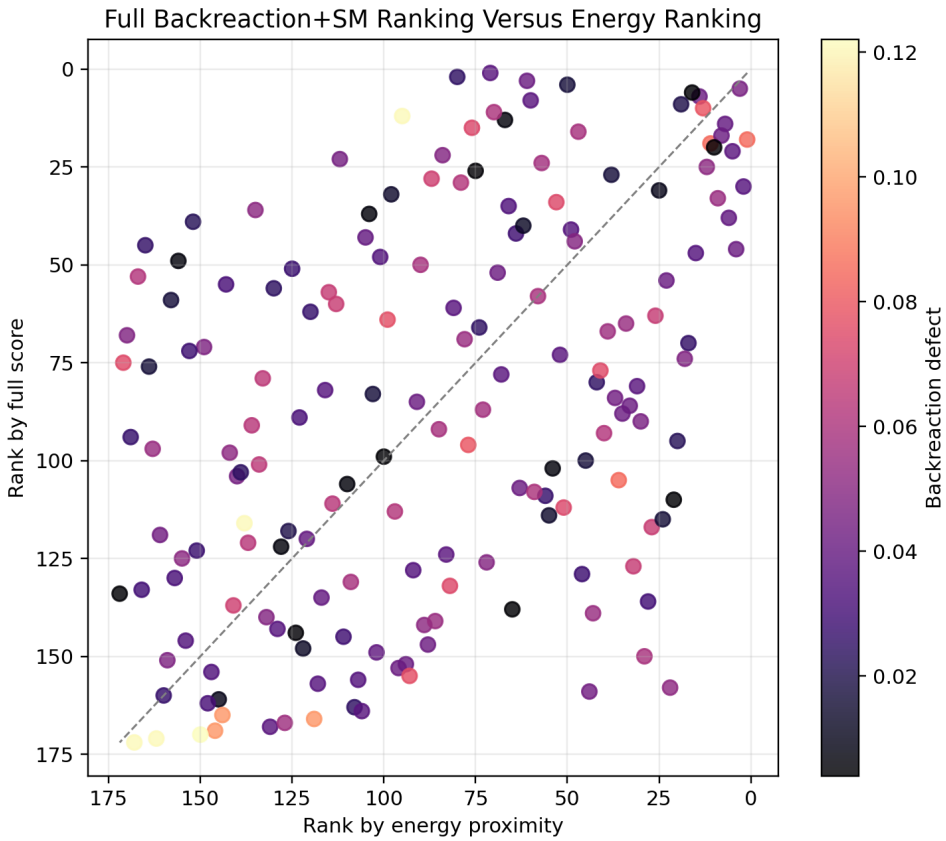

Figure 4: Backreaction and Standard-Model-sector bridge. The full structural ranking differs strongly from energy-only ranking, with top-12 overlap equal to one candidate in the tested ensemble.

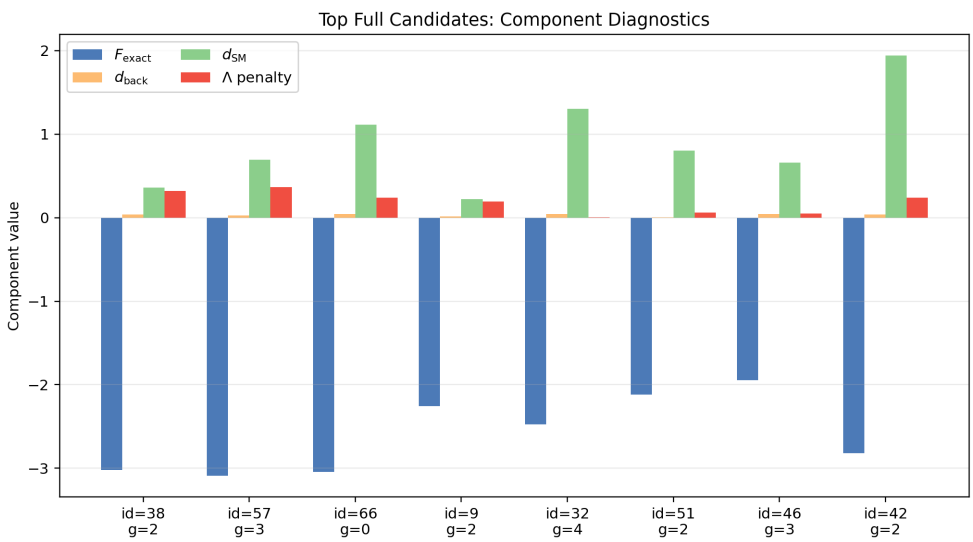

Figure 5: Component diagnostics for the backreaction/SM bridge. The figure decomposes the structural ranking into obstruction sectors, illustrating how different defects contribute to candidate selection.

8 Interpretation

Across the benchmarks, the same qualitative pattern appears:

Low energy is not the same as high structural support.

This is precisely the motivation for a structural selection layer. Energy or vacuum energy remains important, but it does not exhaust consistency, tadpole closure, moduli stability, backreaction control, or phenomenological sector coherence.

The benchmarks do not prove that MAAT selects the real vacuum. They show that a multi-sector structural functional can produce a nontrivial ranking that differs from energy-only criteria and can be evaluated on reproducible ensembles.

9 Limitations

The present framework has substantial limitations:

- the tests are toy and bridge ensembles, not full compactification scans;
- the 10D sector is represented by operational proxies, not complete ten-dimensional supergravity solutions;
- branes, orientifolds, Bianchi identities, warping, and local backreaction are not solved as full field equations;
- Standard-Model-sector quantities are phenomenological proxies, not a constructed chiral Standard Model vacuum;
- the reference measure μ_{string} is ensemble-dependent;
- sector weights λ_a are effective benchmark weights unless derived or calibrated in a specified ensemble;
- the framework does not solve the cosmological-constant problem or prove a unique string vacuum.

These limitations are part of the proposal's current status. The paper defines a ranking architecture and demonstrates that it is computable in small benchmarks; it does not claim finality.

10 Reproducibility

The code, JSON outputs, and figures for the string-landscape benchmarks are available in the public repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/string_landscape_selection

The main scripts are:

```
python3 structural_selection_10d_tadpole_toy.py
python3 structural_selection_iib_kklt_bridge.py
python3 structural_selection_iib_period_kklt_bridge.py
python3 structural_selection_iib_exact_period_kklt_bridge.py
python3 structural_selection_iib_backreaction_sm_bridge.py
```

Each script writes a JSON result file and a plot directory. The numerical outputs quoted in this paper are taken from those generated JSON summaries.

11 Conclusion

We proposed a MAAT-based structural selection layer over the string landscape. The framework keeps string theory as the dynamical base and adds a multi-sector structural ranking over admissible or approximately admissible backgrounds.

The central formula is the landscape functional

$$F_{\mathrm{MAAT}}^{\mathrm{landscape}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad \Gamma_a[X] = \frac{1}{1 + d_a[X]}.$$

In the tested toy and bridge ensembles, structural ranking differs from energy-only ordering and often favours candidates with lower tadpole, stability, or phenomenological-sector obstruction.

The result is not a solution of the landscape problem. It is a reproducible phenomenological framework for making the question more operational:

> *Among dynamically admissible string backgrounds, which ones are structurally best supported?*

References

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